

NAME :- SREEDEEP GHOSH

Project Name :- AI BASED ELDERLY AID

The “AI-Based Elderly Assistance” project provides a new type of voice-controlled, multilingual alarm system for elderly individuals with cognitive, physical, or visual impairments. The proposed new alarm system uses a Raspberry Pi IoT (Internet of Things) platform. The new alarm system uses Natural Language Processing (NLP), speech-to-text, uses Google Text-to-Speech (gTTS) and uses Telegram API to provide a voice-controlled daily companion to the target audience without any screen. The proposed alarm system provides voice controlled scheduled alarm, smart reminders, and SOS emergency tracking alerts with the simple press of a button support independent living, safety, and access. Key features of the alarm system include: multilingual (voice) recognition and processing, alarm processing real time, GPS emergency location tracking, and lightweight module architecture (does not need a visual GUI). Voice is scheduled with the Gemini AI API, and the Telethon library allows the caregiver to respond with SOS emergency SMS, audio, and live coordinates via Geocoder. The system was validated in a simulation setting and demonstrated to be very usable for the target audience, particularly people with disabilities occasioned by natural aging.

NAME :- ROHIT YADAV

Project Name :- ALUMINIX

The Aluminix platform integrates AI to improve the connection between students and alumni by making the platform more intelligent and personalized. An AI-based mentor–mentee matching system recommends suitable mentors to students by comparing their skills, domains of interest, and career goals with the expertise of alumni mentors. This is implemented using a content-based recommendation approach where profile information is converted into text vectors and similarity is calculated using cosine similarity. The system then ranks mentors and suggests the most relevant ones to the student. In addition, a personalized post recommendation system is integrated to show users posts that match their interests, skills, and previously interacted content. This ensures that each user receives a customized feed instead of generic posts. These AI features enhance user engagement, simplify mentorship discovery, and make the platform more efficient for networking and career guidance.

Two research paper one is published and one is yet to

1) Deep Learning Techniques for Medical Image Segmentation-A survey

2)An Enhanced Attention-Based U-Net
for Segmenting Liver Tumours in CT Scans.

NAME :- SAYANTAN HAZRA

Project name :Integrating Sentiment and Emotion Awareness into Semantic Recommendations with Minimal Embedding Augmentation

Traditional text-based recommendation systems rely on semantic similarity from pretrained language models but often ignore sentiment and emotional tone, which strongly influence perceived relevance in social media content. We propose a modular framework that augments semantic text embeddings with sentiment polarity and continuous emotion representations derived from pretrained affect models. A Sentiment-Emotion Block (SEB), consisting of a sentiment scalar and a PAD-based emotion vector, is concatenated with sentence embeddings to create affect-aware representations. The framework is evaluated on a corpus of 27,480 tweets using a composite metric (EmSEB) that balances semantic relevance and affective alignment.

The results show that minimal affective augmentation can significantly improve recommendation quality. The best configuration, Em1SEB.86, increases embedding size by only 1.04% while improving the EmSEB score by 22.75%, driven by strong gains in sentiment (+47.78%) and emotion (+103.94%) alignment, with only a modest reduction in semantic similarity (−13.20%). affective signals can enhance emotional coherence without overwhelming semantic relevance.

Name : INDRAJIT MONDAL

Project Name : Polyp Segmentation

Early detection and assessment of polyps play a crucial role in the prevention and treatment of colorectal cancer. Polyp segmentation provides an effective solution to assist clinicians in accurately locating and segmenting polyp regions. In the past, people often relied on manually extracted lower-level features such as color, texture, and shape, which often had problems capturing global context and lacked robustness to complex scenarios. With the advent of deep learning, more and more medical image segmentation algorithms based on deep learning networks have emerged, making significant progress in the field. This paper provides a comprehensive review of polyp segmentation algorithms. We first review some traditional algorithms based on manually extracted features and deep segmentation algorithms, and then describe benchmark datasets related to the topic. Specifically, we carry out a comprehensive evaluation of recent deep learning models and results based on polyp size, taking into account the focus of research topics and differences in network structures. Finally, we discuss the challenges of polyp segmentation and future trends in the field.

Name : SASWATA MONDAL

Project: SyntaxHut (Competitive Programming Solutions Platform)

Developed a scalable Python backend managing solutions of 2,800+ coding problems, improving search performance by 20%. Built autonomous AI agents using GitHub Actions for automated problem ingestion, solution generation, and continuous deployment, reducing manual workload by 70%. Implemented AI-driven enrichment pipelines for explanations, metadata, and context with a dynamic LLM-routing system. Created a real-time interactive AI chatbot for debugging and code assistance. Deployed the platform on Vercel + Cloudflare CDN, optimizing DNS, SSL/TLS, caching, and SEO, resulting in 212% traffic growth and 213% page-view increase in 30 days.

Paper Title: Attention-Guided Knowledge Distillation for Real-Time Sign Language Recognition on Resource-Constrained Devices

This work introduces an attention-guided knowledge distillation framework combining spatial attention transfer, adaptive sample reweighting, and layer-wise distillation to compress YOLOv8-Medium for real-time sign-language recognition. On a 36-class ASL dataset, the method achieves **99.50% mAP@0.5** and **98.91% mAP@0.5:0.95**, outperforming standard KD while remaining nearly identical to teacher performance. INT8 quantization reduces model size by **45.4%** (5.97 MB $\rightarrow$ 3.26 MB). The distilled model runs in real-time at **54.9 FPS** on an RTX 4050, with projected **15–25 FPS** on mobile and **5–10 FPS** on edge devices. Overall, the approach preserves accuracy while enabling deployment on resource-constrained platforms.